

Black-Scholes implied volatility as a range: exact quotes, rounding cells and auditable bounds

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1. Result and scope

A rounded option price need not identify one volatility. We provide a small Python research utility that maps an exact price interval or a binary64 rounding cell to an outer enclosure of the compatible Black-Scholes-Merton (BSM) volatilities. Its output distinguishes an empty set, zero volatility, unbounded ranges, expiry and insufficient computational budgets. It retains exact rational endpoints and dyadic witnesses for definite comparisons.

For a call with $S=200$, $K=100$, $T=1$ and $r=q=0$, the binary64 observation 100 is compatible with every volatility from zero to approximately **8.9407687355%**. We enclose the upper endpoint in an interval narrower than 10^{-24} in sigma. This refines the illustrative information-loss example in our [preceding working paper](#).

The final fresh confirmation has **64/64 scenarios passing all declared gates**: 56 resolved outcomes and eight deliberately insufficient budgets handled explicitly. Earlier confirmation runs scored **158/160 and 82/96** because of test-generator/reference errors; those results are retained, not relabelled. The candidate remained unchanged throughout all three runs.

This is an engineering artifact using standard BSM mathematics and Arb enclosures. It is not a new market model, priority claim, performance record or independently verified formal proof. Quoted ranges are conditional on exact model parameters and the specified interpretation of the price.

2. Why inversion produces a set

Put $A=S \exp(-qT)$, $B=K \exp(-rT)$. For $T>0$, the call has intrinsic lower limit $L=\max(A-B,0)$ and unattained upper limit $U=A$; for the put, $L=\max(B-A,0)$ and $U=B$. For positive sigma, model Vega is strictly positive:

$$dP/d\sigma = A \sqrt{T} \varphi(d_1) > 0.$$

Thus P is continuous on $\sigma \geq 0$, strictly increasing, with attainable range $[L,U)$. Given an exact quote interval Q , the quantity of interest is

$$\Sigma(Q) = \{\sigma \geq 0 : P(\sigma) \text{ is in } Q\}.$$

Intersect Q with $[L,U)$. An empty intersection means no finite compatible volatility. A sole included price L gives $\{0\}$. Every interior endpoint has a unique inverse, and its open/closed status is inherited. If the quote extends to U , the volatility set has no finite upper bound even when U itself is excluded. At $T=0$, the payoff is independent of volatility: either all nonnegative volatilities are compatible or none are.

For a positive finite binary64 observation p , the real inputs rounding to p are bounded by midpoints with its adjacent representable values. Both ties belong to the cell exactly when p has an even least-significant significand bit. At zero, the nonnegative cell is $[0, 2^{-1075}]$. At the largest finite number, the overflow midpoint uses the virtual successor 2^{1024} and is excluded. These cells assume round-to-nearest, ties-to-even. Decimal quote intervals and binary64 cells are different input contracts.

3. Computation and meaning of the certificate

The implementation evaluates the ordinary unspecialized BSM erfc expressions using python-flint 0.8.0's Arb arithmetic. Exact rational inputs are converted to enclosing balls; exponential, logarithm, square root and erfc retain enclosures. A comparison is used only when the entire price-minus-target ball is below zero, above zero, or exactly zero. Unknown comparisons never become equality. Arb's documented enclosure contract is an explicit dependency; we have not independently verified its native implementation.

Root search doubles an upper trial until it proves a bracket, then bisepts using exact rational midpoints. Monotonicity and the two proved endpoint signs imply root containment. Precision increases along a declared schedule. An uncertain trial during initial bracketing is not a valid finite upper bound: the result preserves a conservative half-line and an unresolved status. After a bracket is established, exhaustion retains that bracket.

The JSON output stores exact rational outer bounds and the brackets locating the exact mathematical endpoints. Endpoint closure flags describe those mathematical endpoints, not rounded bracket endpoints. `resolved=true` means endpoint location meets the requested absolute sigma tolerance; it does not mean the quote uniquely identifies sigma. A finite mathematical range may have an infinite *reported outer bound* if the search budget is exhausted. The separate unbounded field describes the mathematical case.

Resource limits include rational bit size, precision at most 4096 bits, at most 2048 bisections and a sigma search cap. Default endpoint tolerance is 10^{-12} . A single process and `ctx.threads=1` are required because precision context is global. These limits are not a universal runtime or memory SLA.

4. Worked examples

The four post-confirmation illustrations have their own receipt and are not counted as new confirmation. All their retained witnesses pass the separate high-precision control. For the rounded ITM call of 100, the exact upper endpoint lies strictly between these outward decimal bounds:

0.0894076873552443926268368

0.0894076873552443926268378

The underlying rational bracket has width below 10^{-24} and required up to 256-bit arithmetic. Hence the six volatilities 3%, 4%, 5%, 6%, 7% and 8% in v4 are part of a continuum of compatible answers, not six isolated cases.

For the corresponding OTM put, a binary64 observation generated at `sigma=5%` still retains the tiny positive time value. Its returned outer enclosure is contained between 0.04999999999999999769 and 0.050000000000000000312. This contrasts two storage representations of synthetic model prices; it does not imply that real markets quote such tiny prices or that put and call bid/ask spreads are interchangeable.

A simpler call example $S=K=100$, $T=1$, $r=q=0$ and exact price band $[7,9]$ yields sigma approximately $[0.175689675792, 0.226077081290]$. Tightening numerical tolerance cannot remove the width caused by that price band. A fourth illustration near `sigma=0.5` forces escalation from 64 to 512 bits. The main confirmation corpora needed at most 128 bits; precision escalation is tested separately, not implied by their counts.

5. Evidence, including the failed checks

Every confirmation corpus and its candidate, generator, runner, reference, protocol and dependency hashes were fixed locally before viewing candidate outputs on that corpus. The cases are deterministic synthetic tests, not market samples, random population estimates or external blind evaluation. The primary implementation hash is 2519288a7ee616274ad7f87f18ec2d6021e4407b4bc7272b1674bcb7b252985a.

Frozen run	Scenarios	All original gates pass	Retained definite witnesses
IV v1	160	158	1108
IV v2	96	82	673
IV v3	64	64	417

In v1, converting an mpmath value directly to float selected a neighbouring subnormal rounding cell in one generator case. The discrepancy persisted at 180, 260 and 360 decimal digits; Fraction-mediated conversion and a 1024-bit Arb check diagnosed it. Another scenario incorrectly expected boundary ambiguity, while Arb proved a sign; two control convergence checks also suffered cancellation of a tiny intrinsic value. The frozen failures remain visible. No maintainer defect or novelty claim is made here.

The v2 control used expm1 for tiny intrinsic differences, but unnecessarily passed exact zero-carry differences through log/exp. Tiny oracle rounding errors then failed exact, zero-width witness checks in 14 scenarios. The v3 control preserves exact rational S-K in those cases. Rechecking the earlier 1781 definite witnesses with this control passed, explicitly as **post hoc diagnosis**, not as replacement confirmation. No solver change was needed.

The final 64 new scenarios are disjoint from both earlier corpora: 32 rounded quote recovery cases, 16 rounded boundaries, eight explicit budget limits and eight expiry cases. All 417 retained definite witnesses have valid dyadic signs and contain the separately computed 260-digit value, with the declared 180/260 convergence checks. All 32 latent sigmas lie in the returned outer ranges. The eight budget cases remain unresolved as intended. Development has 24 solver test methods plus three reference-identity methods; v1 also passes eight parity/scaling checks. Failed development logs remain.

6. Reproduce and inspect

Use Python 3.12 with python-flint==0.8.0 and mpmath==1.3.0; no GPU is required. The distributed package contains original source and evidence, not native dependency binaries. Set numerical thread counts to one. From the extracted release, using fresh output paths:

```
python -m unittest discover -s iv-tests -v
python -m unittest discover -s iv-tests-v3 -v
python -m lab.run_iv_v3 --portable --output /tmp/iv-replay.json
python iv_range.py --spot 200 --strike 100 --time 1 --rounded-price 100
```

The portable runner checks source/input hashes and dependency versions; platform-specific native library hashes may differ. The original native hashes are retained for audit. Replaying lab.run_iv and lab.run_iv_v2 reproduces their original failed gates and intentionally exits nonzero. Reproduction on this host is not independent human review.

Code and complete evidence: [GERO Numerical Observatory](#), interval-IV v5.

7. Limitations, related work and authorship

The set inversion follows elementary continuity and monotonicity; enclosure arithmetic and bracketing are established methods. Arb is described by Fredrik Johansson, *Arb: Efficient Arbitrary-Precision Midpoint-Radius Interval Arithmetic*, [arXiv:1611.02831](https://arxiv.org/abs/1611.02831). [python-flint documentation](#) describes its real-ball operations; experiments pin version 0.8.0, while the current documentation displays 0.9.0. We do not claim to have reproduced the benchmarks in Wolfgang Schadner's [explicit IV representation](#). That work studies a representation of the inverse; our focus is retaining uncertainty in the observed price. No superiority comparison is made.

Uncertain spot, rates, dividends, time, exercise features, volatility smiles, transaction costs and model misspecification are outside this API. A bid/ask interval is not automatically a statistical confidence interval. Production integration, specialist solver comparisons and external review remain open.

Khamit Kadyrbekov initiated and directs this programme. Daniyal Kadirbekov is coauthor and designated lead for continuing software development; this does not assert independent implementation or validation by him. OpenAI Codex materially assisted derivation, implementation, experiment design, execution, diagnosis and drafting. No university affiliation, endorsement or independent human peer review is claimed. Computation was sequential on CPU with one numerical worker. Original text/data: CC BY 4.0; code: MIT.